Dominance can move when corridors move.

Centre for Blockchain Technologies

The Making of Dominant Vehicle Currencies

Corridors, local bridges, and liquidity provision

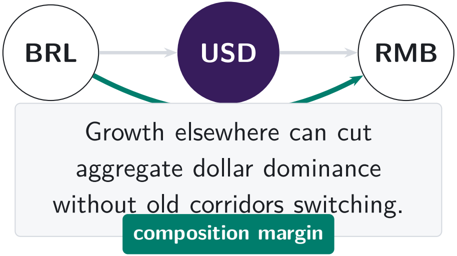
Jiahua Xu

Dominance changes when trade corridors change

TRADITIONAL-FINANCE INTUITION

A treasury desk keeps one thin dollar corridor while aggregate dollar share falls.

Growth elsewhere changes the denominator: route allocation, not only within-pair substitution.



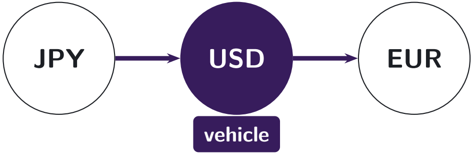
Note: The clearing example is an analogy for the denominator problem. The empirical object in this paper is the intermediary asset observed inside DEX pool routes.

Vehicle currencies are routes

FX ANALOGY

Yen–euro may trade through dollars.

Pair data see two legs, not the customer route.



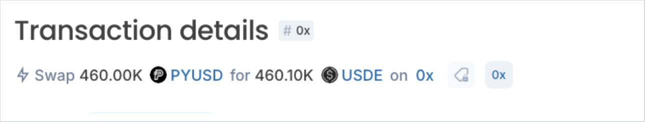
Pool logs reveal the path.

DEX data make the hidden route observable.

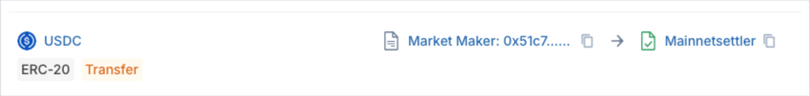
Note: Traditional pair-level FX records show turnover in each bilateral pair, but not whether two legs belong to one customer route. The DEX setting preserves the transaction-level pool path, so the intermediary asset is observed directly.

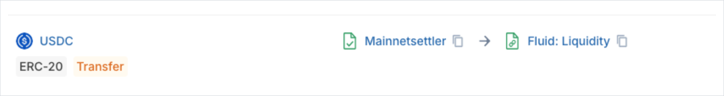
The PYUSD swap reaches a USDC-funded pool route

USER INSTRUCTION RECORDED BY THE EXPLORER



THE DECISIVE SETTLEMENT TRANSFERS





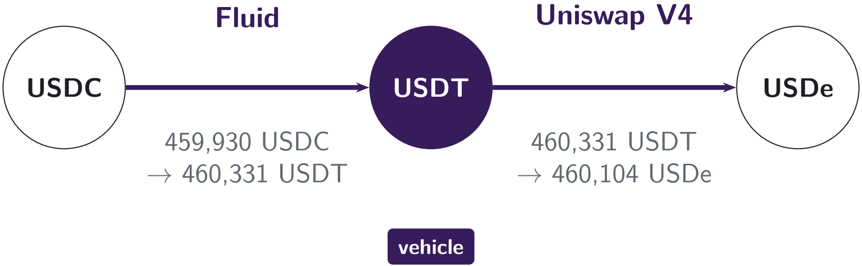
The instruction is PYUSD $\rightarrow$ USDe. The maker supplies the USDC used by the Fluid leg, which begins the connected pool route.

Note: The explorer screenshot and transfer trace refer to transaction `0xbda1d07f...9bf67ad9` on 10 January 2026.

Inside the pools, USDT links USDC to USDe

ORDERED TRANSFERS AND ROUTE RECONSTRUCTION

User instruction: PYUSD → USDe ⟹ Pool route: USDC → USDT → USDe



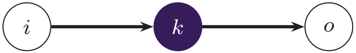
The observed ultimate trade is USDC → USDe. Its atomic trades are USDC → USDT and USDT → USDe; USDT is the vehicle that links them.

Note: The route is the connected pool-swap component of transaction `0xbda1d07f...9bf67ad9` on 10 January 2026.

Vehicle dominance is the share of indirect trade carried by an asset

MEASUREMENT

- Dominance is the share of indirect trade intermediated by asset k.
- Count share treats each route equally; value share weights routes by dollar value.
- Network measures capture how broadly k connects ultimate pairs.

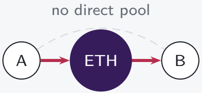


Count-weighted dominance
Share of indirect routes that pass through k

Value-weighted dominance
Share of comparable routed value that passes through k

Architecture changes routing opportunities

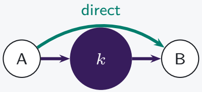
V1 mandatory hub



Only ETH–token pools: token-to-token execution must pass through ETH.

Hub use is imposed

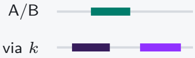
V2 pair choice



Arbitrary pairs add a direct route; vehicle routes remain feasible.

Hub use becomes a choice

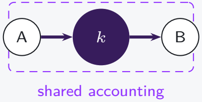
V3 range choice



LPs choose ranges and fees; active depth can favor direct pairs or vehicle spokes.

Opportunity shifts by pair

V4 net settlement



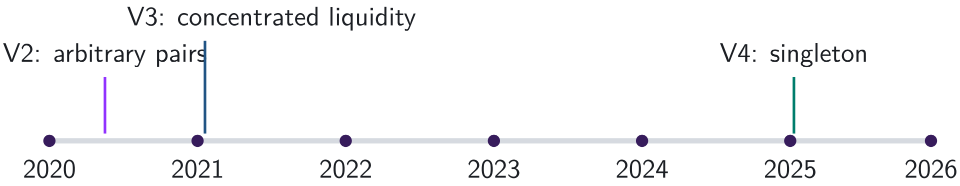
Two pools still execute; intermediate token movement may be netted.

Settlement can diverge from route use

Architecture feasible rules **Exposure** realised pool and route use **Calendar date** timing only

The route sample and architecture supplement answer different questions

DATA



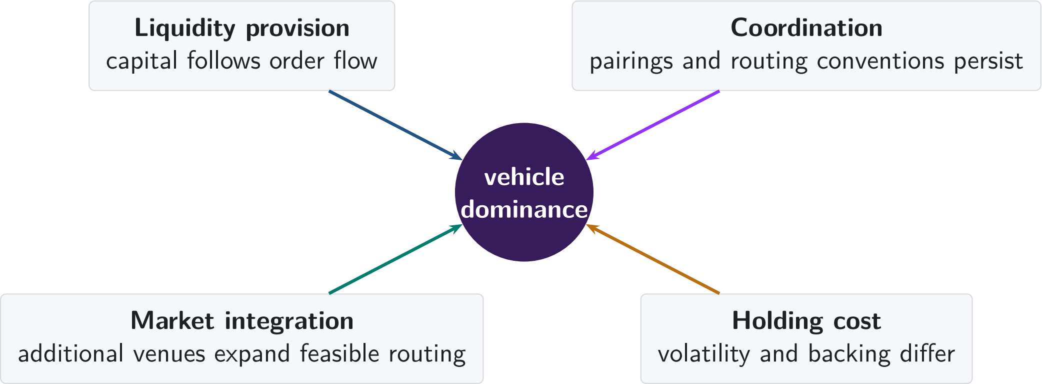
Main route sample

Atomic trades (swap legs) from multiple protocols are linked within each transaction. An atomic pair may be a two-asset pool or one asset pair inside a multi-asset Curve or Balancer pool.

V1/V2 architecture supplement

Uniswap V1 forced-ETH paths and V2 pair formation are analysed separately. They answer protocol-design questions and do not define the main route perimeter.

Why might one asset become the dominant vehicle?



Native-asset pairing persists after it becomes optional

V1: PROTOCOL RULE

217,003 token-to-token trades pass through ETH

Every token pool pairs with ETH, so the architecture imposes the vehicle.

$\Longrightarrow$

V2: MARKET CHOICE

95.5%

of single-leg V2 trades use a WETH pool in 2026

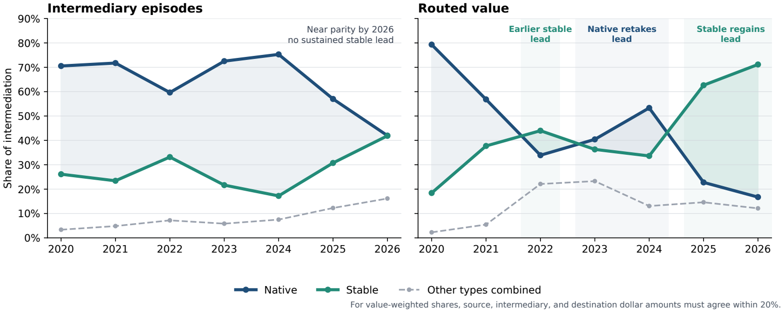
97.9%

of pairs first traded in 2026 include WETH

V2 converted ETH pairing from a protocol rule into a market choice. Native-asset concentration remained high through pair formation and realised trading.

Note: V1 counts linked token-to-ETH and ETH-to-token swaps with matching ETH quantities in one transaction. V2 shares use single-leg Uniswap V2 trades and newly traded pairs; they describe that venue rather than market-wide vehicle dominance.

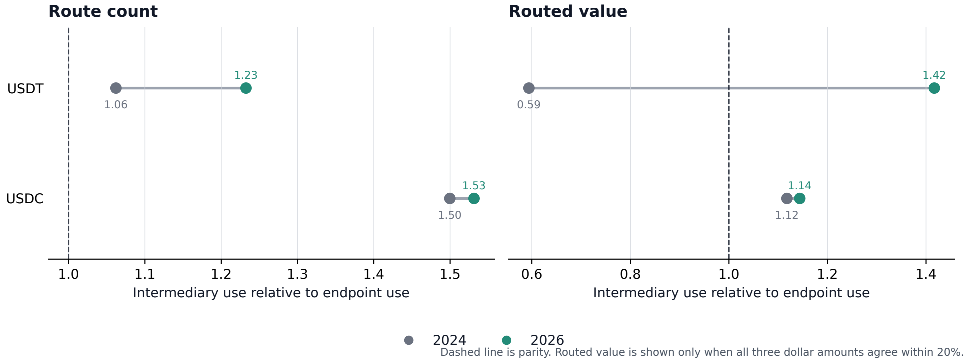
Stablecoins regain the lead in routed value



The aggregate share moves because the network of traded ultimate pairs reorganises around stablecoins. Holding the ultimate pair, the calendar date, and the route scope fixed, stablecoin share changes by $+0.2$ pp (SE 0.8 pp) by count and -1.3 pp (SE 2.2 pp) by value.

Note: The route universe is exact two-leg routes (two atomic trades) with an observed intermediary. The left panel counts one intermediary episode per route, hence per ultimate trade, and divides each asset type's episodes by all intermediary episodes. The right panel divides each type's routed value by total routed value among routes whose source, intermediary, and destination dollar amounts agree within 20%. The 2026 observation covers January–June; Other combines imported, liquid-staking, and residual assets. Matched-market estimates weight by route count or value and cluster by ordered ultimate pair and calendar date.

USDT drives the rise in stablecoin intermediation



USDT crosses parity by routed value and extends its count advantage; USDC changes little.

Note: Excess use is intermediary share divided by endpoint share on the same route perimeter. The comparison uses native assets, USDC, and USDT and the same January–June dates in 2024 and 2026. Descriptive accounting: see Appendix A9; the identity carries no sampling interval.

The regression compares currencies within the same day

$$I_{a,t} = \alpha + \beta D_{a,t} + \mathbf{A}_a'\theta + \mu_t + \eta_a + \varepsilon_{a,t}$$

- $I_{a,t}$: currency share of intermediary episodes
- $D_{a,t}$: own route-endpoint demand share
- $\mathbf{A}_a$: native, stable, and other classes
- μ_t: date effects absorb the calendar
- η_a: currency effects absorb persistent identity
- SE clustered by date and currency

Pooled $\rightarrow$ date effects $\rightarrow$ own demand $\rightarrow$ currency effects.

Note: The outcome and demand regressor are percentage-point shares of the same day's route universe. The specification ladder is reported in the next frame. This is descriptive within-day identification, not a causal demand shock.

The cross-section holds when the calendar is absorbed

	Native	Stable	Demand
Pooled	$+34.56$ pp	$+2.42$ pp	—
$+$ day effects	$+34.55$ pp	$+2.42$ pp	—
$+$ own demand	$+0.05$ pp	$+0.85$ pp	1.587
$+$ currency effects	—	—	0.985

Day effects leave the class premiums where they were. At the same trade demand the native premium is $+0.05$ pp (0.56 pp) and the stablecoin premium $+0.85$ pp (0.50 pp).

Note: 1,828,862 currency-days, 304,572 currencies, 2,259 days. Premiums are percentage points of the day's intermediary episodes against the residual unclassified bucket; demand is the pass-through of the currency's own route-endpoint share. Standard errors cluster on date and currency.

WETH-linked ultimate pairs account for most of the count rotation

WETH

WETH is the source or destination

WETH cannot also be the intermediary; stable share is mechanically 100%.

All matched trading-pair groups

$+21.1$ **pp**

95% CI $[+18.2$ pp, $+24.0$ pp$]$

WETH-linked route-count mass: 26.6% $\rightarrow$ 46.8%

$\xrightarrow{\text{exclude WETH endpoints}}$

Trading-pair groups without WETH endpoints

$+3.7$ **pp**

95% CI $[+2.3$ pp, $+5.2$ pp$]$

79,347 matched groups

Within the same trading pair, calendar position, and route scope: all groups $+0.2$ pp $[-1.3$ pp, $+1.7$ pp$]$; without WETH endpoints $+0.3$ pp $[-1.9$ pp, $+2.6$ pp$]$.

The count increase falls from $+21.1$ pp to $+3.7$ pp without WETH endpoints. A WETH-linked pair admits only a stablecoin intermediary in this comparison, so the count rotation largely records where trading happens rather than which intermediary a given trade selects.

Note: Exact two-leg native-WETH-versus-stablecoin routes on 181 common month-days. Intervals use 30-day Newey–West covariance on 362 endpoint-year daily observations; 2025 is excluded and actual dates preserve the gap. Within-group WLS fixes the ultimate pair, month-day, and route scope and clusters by pair and date. Feasible routes are not fixed; all comparisons are descriptive.

The routed-value rotation extends beyond WETH-linked ultimate pairs

Without WETH endpoints		Routed-value reallocation		Change within groups
$+21.5$ **pp**	$=$	$+23.5$ **pp**	$+$	-2.0 **pp**
95% CI [$+11.8$ pp, $+31.1$ pp]		95% CI [$+15.0$ pp, $+32.1$ pp]		95% CI [-9.8 pp, $+5.7$ pp]
strict-support stablecoin share of routed value Full matched sample: $+27.2$ pp [$+23.4$ pp, $+30.9$ pp]		Value shifts toward trading-pair and route-scope groups with higher stablecoin shares.		Stablecoin share changes within the same trading-pair and route-scope group.

Exact midpoint identity on each of 181 common calendar days; 2024 versus January–June 2026.

After removing WETH endpoints, value moves toward trading-pair and route-scope groups in which stablecoins already carry more intermediation. The change within those groups is small and imprecise.

Note: Routed value requires source, intermediary, and destination dollar amounts to agree within 20%. The daily midpoint identity is exact. Intervals use the same endpoint-year calendar-HAC covariance as the preceding frame; paired-month-day and bandwidth sensitivities remain in the source. Mukhin (2022), Gopinath and Stein (2021), and Carletti et al. (2021) guide the composition framing, not identification.

Three margins move an aggregate vehicle share

1 Used more often within a pair

USDC → WBTC trades in both years

Stablecoin share of its intermediary use

10.5% → 35.9%

Adds $+0.05$ pp to the aggregate

Margin total: -0.1 pp

2 Trading moves toward pairs that already use it

USDT → WETH, stablecoin-routed in both years

Share of all routed activity

0.36% → 4.52%

7,447 routes become 54,112; adds $+2.17$ pp

Margin total: $+8.6$ pp

3 Newly traded pairs route through it

USDS → WETH, absent from the base year

Routes observed

0 → 3,390

All stablecoin-intermediated; adds $+0.13$ pp

Margin total: $+21.0$ pp

The first margin is the one vehicle-currency theory describes, and it did not move. The third nets to $+17.8$ pp because 48.0% of routes run on pairs alive in only one year.

Note: Pooled route-count share among native and stablecoin intermediaries, 2024 against 2026 on 181 common calendar days. Each panel names the largest contributing ordered pair with two canonical symbols; unlabelled assets carry most of the third margin. Contributions are an exact allocation of the aggregate change and are descriptive.

Vehicle choice is sticky from market birth

5.0%

stable share at entry, 2024

25.1%

stable share at entry, 2026

$+97.0$ **pp**

30-day stable-born gap

Entrant state sticks: stable-born pairs stay at 97.5%, native-born at 0.5%. High-value entry scopes are sharper: 97.8% stable in 2026, up from 48.4%. In no-direct corridors, $+10$ pp entry stable share predicts $+8.9$ pp at 120 days. Named identity sticks too.

Note: Entrants are first-observed ordered pairs in matched January-June windows. Path, value, and future-activity specifications use non-WETH entrants with standard controls; future activity excludes entry day. The 120-day direct-route split has 512,020 no-direct entrant rows and 181 direct-route-present rows. Identity specification uses 6,087 stable-entry candidate rows with candidate fixed effects. Descriptive association only.

USDC kept the vehicle role while stable capital stepped back

67.9%

USDC stable-route share during stress

$+11.4$ **pp**

stable basket route-share change

-2.0 **pp**

stable basket capital-share change

Routes did not flee USDC during the SVB weekend: its stable-vehicle share moved -0.2 pp while stable-route activity rose 73.6%. LP capital moved the other way: post-window WETH capital share rose $+3.1$ pp and USDC capital share fell -3.2 pp.

Note: Daily route counts and V2 deposited-capital shares from the five-candidate route-capital panel. Windows are the 30 calendar days before March 10, the March 10–13 stress interval, and the 30 calendar days after. Route identity is within stable candidates; route and capital changes are within the five-candidate set. The event mixes peg stress, arbitrage, and demand shocks, so this is descriptive, not a treatment effect.

Stable vehicles turn on at the market edge

-2.7 **pp**

stable turn-on per log baseline routes

$+27.1$ **pp**

stable turn-on per baseline direct-share unit

-1.1 **pp**

stable-majority switch per log baseline routes

Thin/direct/complex corridors turn on. In mixed risk sets, reach predicts share. Stable remains negative, but 2026 USDT gets a $+88.2$ pp issuer premium after reach controls.

Note: Transition screen: pair-day-scope rows, month-day FE, harmonic weights, two-way CR1. Risk-set rows are observed native-stable candidate sets, not feasible alternatives; reach is leave-one same-day or prior-30-day pair-scope count. Issuer rows add USDC/USDT indicators to the same design. Market-size rows are descriptive; driver screen only.

Native-only markets later add stable routes

2.0%

30-day stable turn-on, bottom size bin

24.5%

30-day stable turn-on, top size bin

41.4%

30-day stable turn-on, oldest bin

Among active native-only pair-days, stable-vehicle use over the next month is much more common in thick and older markets. Month-day FE slopes stay positive for market thickness ($+3.5$ pp) and pair age ($+1.6$ pp).

Note: Rows are 1,690,478 active ordered pair-days across 620,849 pairs with positive native-or-stable vehicle use and zero stable vehicle use on the origin day. Outcome is any stable-vehicle use over the next complete 30 days. Quantile bins and FE estimates are weighted by origin market-route count; FE absorbs month-day and clusters by ordered pair and date. Descriptive hazard, not causal timing.

Extra-hop execution became cheap enough

152.7%

more gas than direct, stable-vehicle median in 2026

$0.08

fixed extra-hop toll at 2026 medians

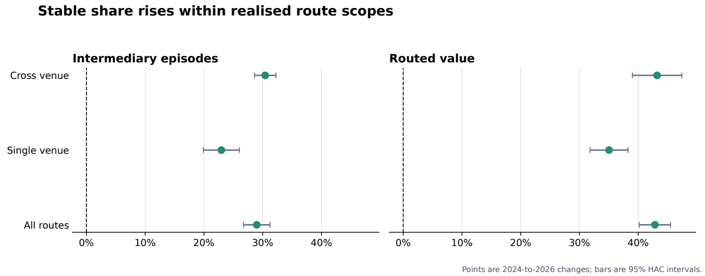
$762

notional where extra gas is 1 bp

In gas units, stable-vehicle routes still cost more than direct routes: the 2026 median extra toll is 233,640 gas. But median stable-vehicle notional fell from $832 to $82, while the median fixed toll fell from 44.3 bp to 16.2 bp.

Note: Executed-route receipt gas only. Dollar tolls evaluate the stable-vehicle extra-gas median against the same year's median gas price and WETH price. At route-specific gas prices, the share of stable-vehicle routes with fixed extra-hop toll below 25 bps rises from 39.5% to 55.7%; the 2026 ten-basis-point threshold is $76, close to the observed stable-vehicle median notional of $82. This is execution-friction evidence, not an all-in quote comparison.

Stablecoin use rises within each venue scope



Neither venue mix nor path length absorbs the rotation: two-leg routes gain $+25.4$ pp (SE 1.05 pp), longer routes $+16.3$ pp (SE 1.86 pp), and the weakest venue-by-length cell still rises by $+17.0$ pp. Nor does the pricing rule or automated path search. Within a day the venue gap is a value phenomenon: cross-exchange one-intermediary routes carry $+14.18$ pp more stablecoin value share (SE 3.69 pp), against $+3.73$ pp on $[-2.2, +9.7]$ by count.

Note: Changes compare the same January–June dates in 2024 and 2026; bars show 95% calendar-HAC intervals. Excess-use ratios and the within-day venue gap span the full sample calendar.

Routes follow local bridges, not only popular vehicles

84.1%

route share captured by deepest bridge

75.5% $\rightarrow$ **86.5%**

top-bridge share, 2024 to 2026

$+6.74$ **pp**

route-share slope after candidate-reach controls

At market birth, $+1$ log point of local bridge depth predicts $+5.39$ pp route share and $+5.32$ pp selection. In the full panel, weak-leg depth remains $+6.57$ pp after reach controls; leg imbalance predicts -5.42 pp.

Note: Five candidates; 137,003 candidate rows, 2,923 pairs, 363 days. Birth rows use 835 candidate rows and retain zero local depth. Depth is the smaller prior-capital bridge leg. Reach uses same-day or prior-30-day routes. Stable-issuer rows use 12,048 DAI/USDC/USDT cells. Association only.

Stable vehicle markets show feedback

$+0.115$

later stable bridge-depth growth from a 10 pp higher current route share, 120 days

$+2.46$ **pp**

later stable route-share growth per log point of current local depth, 120 days

Route use predicts later bridge deepening, and depth predicts later route-share gains, on the same local source-candidate-target bridge. The feedback margin is stronger for stable candidates than for the pooled five-candidate set.

Note: Provisional feedback screen over 18,304 120-day rows and 340 ordered pairs. The stable 120-day log-depth coefficient is a 10 percentage point route-share effect, with SE 0.012. The stable 120-day route-share coefficient has SE 0.29 pp. Regressions control for current depth or current route share, absorb candidate and origin-date effects, weight by five-candidate route count, and cluster by ordered pair and date. Association only, not provider-flow causality.

Provider capital stays native-heavy

100.0%

WETH leads deposited capital

81.3%

stablecoins lead excess use

$4.2\times$

2026 route share over capital share

Same candidate-date: V4 action, reported TVL, and pool-footprint responses exceed V3 ($+0.320$ log pts, $+0.637$ log pts, $+0.275$ log pts). Gross/add flow is flat; remove turnover is $+0.046$ log pts. Flash value shifts narrow/medium while actions move wide.

Note: Five-candidate V2 capital panel. V3 rows use 11,195 mint/burn records and reported pool TVL; V4 rows use filtered candidate-token flow, flash-LP, reported pool TVL, and stable-gap interactions. Protocol contrasts use candidate-date and protocol effects. Reported TVL is candidate-linked full-pool TVL, not reconstructed capital, LP inventory, or causal protocol treatment.

Deposited capital neither leads vehicle use nor follows it

LIQUIDITY FIRST

Deposited capital in vehicle-linked pools at t

↓

Vehicle share at $t + \tau$

A one log-point higher capital stock predicts a next-day episode-share change of 0.00 pp (SE 0.04 pp); at thirty days -0.64 pp (SE 0.26 pp).

VEHICLE USE FIRST

Vehicle share at t

↓

Deposited capital in vehicle-linked pools at $t+\tau$

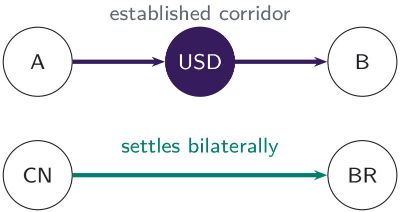
A one-point higher episode share predicts a next-day capital change of -0.03% (SE 0.03%); at thirty days -0.32% (SE 0.27%).

No use/capital measure pair is positive in both directions at any two of the one-, seven-, and thirty-day horizons; the smallest Holm-adjusted p is 0.17. The only strong longer-horizon pattern runs the other way: over the following 120 days, higher-capital pool-days lose episode share (-2.21 pp per log point, SE 0.39 pp).

Note: Daily panel of the five vehicle candidates on the full 2,238-day V2 calendar with candidate and date fixed effects; standard errors use 30-day calendar HAC on date-aggregated scores, and Holm adjustment is within direction; a month-block bootstrap and candidate-date clustering agree. Deposited capital is a reserve stock; provider flows and concentrated-liquidity depth are measured separately. Predictive associations only, not causal feedback.

The same arithmetic governs any aggregate currency share

A FAMILIAR SETTING



China and Brazil have built clearing arrangements for settling their bilateral trade in renminbi. The established corridor above is unaffected.

THE SHARE CAN FALL WITH NO CORRIDOR SWITCHING

A currency's share of all settled trade falls when trade grows fastest along corridors that never adopted it, even if every corridor that uses it today keeps using it tomorrow.

Margin 2 is trade reallocating toward corridors that already avoid the incumbent.

Margin 3 is a corridor that had no meaningful volume opening on other terms.

Our routes measure the same two margins directly, because the transaction log records the whole path rather than its legs.

Note: Illustrative comparison only. Renminbi clearing arrangements in Brazil follow the memorandum of understanding between the People's Bank of China and the Banco Central do Brasil. No currency share of trade settlement is measured here, and no causal or policy claim is made about either setting.

The market changed, not the trade

SHARE OF INTERMEDIATION CARRIED BY STABLECOINS

as the comparison holds more of the market fixed

Across the whole market	Within pairs traded in both years	Same pair, same day, same route scope
$+25.7$ **pp**	-0.1 **pp**	$+0.2$ **pp**
no conditioning at all	theory's own margin	SE 0.8 pp, over 94,260 matched cells

Stablecoins won through the markets that formed and grew, not repeated switching inside established pairs.

The three shares are one measure at three levels of conditioning and are directly comparable. They describe realised route composition; they are not entry or exit effects and do not identify a treatment.

Appendix map

Definitions and data

- A1 Route units and vehicle status
- A2 Asset types and backing regimes
- A3 Sample construction and exclusions

Reconstruction and validation

- A4 Exact transaction state
- A5 Protocol-specific state
- A6 Forced routes in V1

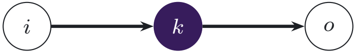
Robustness

- A7 Daily and weekly frequencies
- A8 Count and value weighting
- A9 Observed route endpoints

Mechanisms

- A10 Capital, inventory, and depth
- A11 Coordination and integration
- A12–A13 References

A1. One reconstructed route is one unit

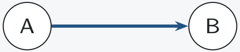


one route unit
two swap legs

- Each linked $i \to k \to o$ route is counted once.
- Asset k is classified as the intermediary on that route.
- Dominance is the share of route units or supported value carried by k.

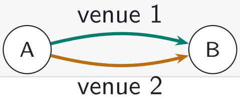
How we reconstruct a route from one transaction

Direct swap
one transaction

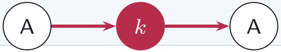


Sequential vehicle route
one transaction

Parallel split
one transaction

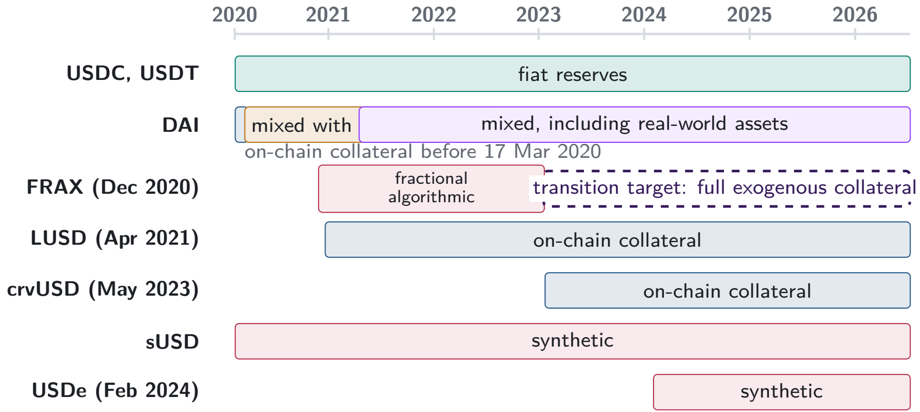


Round trip
one transaction



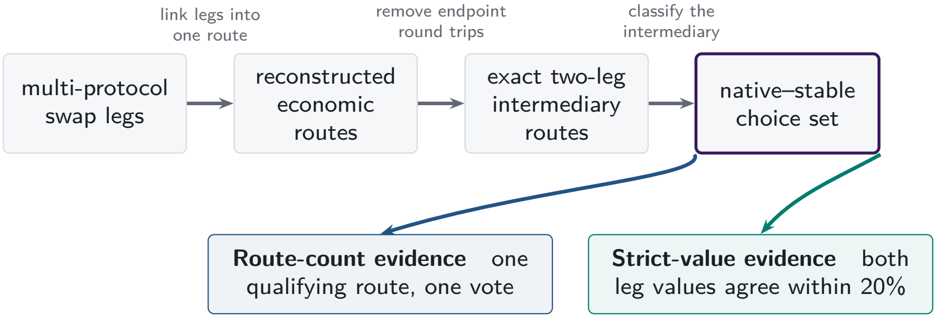
Vehicle choice directs indirect order flow toward particular assets and pools. Transaction-level paths reveal that choice rather than leaving it inside aggregate turnover.

A2. Stablecoin backing cannot be treated as fixed



Note: Bars begin at public launch or the sample start for older tokens. DAI's dated regimes follow the Maker Governance actions that added USDC-A and activated RWA001-A. FRAX's dashed arrow follows FIP-188 and denotes a target, not observed collateral.

A3. One route universe supports two measurements



Box widths do not represent attrition.

The route sample links legs across protocols; one asset pair inside a multi-asset Curve or Balancer pool may supply a leg. The V1/V2 architecture evidence is separate. Dollar support affects value weighting, not route inclusion.

A4. State is reconstructed immediately before execution

1. verified starting state →
2. apply events in blockchain order →
3. recover state before the trade →
4. quote both routes

- Pool events are applied in blockchain order from a verified starting state.
- Each route is quoted using the pool state immediately before the transaction.
- Missing state intervals are excluded, and their share of trading volume is reported.

A5. Each AMM family has a different state vector

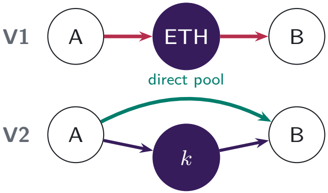
	reserves or active liquidity	ticks	amplification	weights or scaling	rates or oracles	fees	hooks
Constant product	■	–	–	–	–	■	–
Concentrated liquidity	■	■	–	–	–	■	–
StableSwap families	■	–	■	–	■	■	–
Weighted pools	■	–	–	■	■	■	–
V4 singleton	■	■	–	–	–	■	□

The common object is the state immediately before execution. The fields required to recover that state vary with the pricing rule.

■ required state
□ hooks outside comparison

Protocol design changes markets, depth, and settlement

V1 $\longrightarrow$ V2: PAIR CREATION



V1 admits only ETH–token pools. V2 permits any ERC-20 pair, so market creation determines whether direct and vehicle paths coexist.

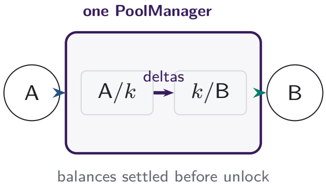
Which edges exist?

V3: ACTIVE LIQUIDITY

LPs choose a price range and fee tier for each pool. Positions covering the current price determine active liquidity in the direct pair and in each vehicle spoke.

Which path offers depth at this size?

V4: SHARED ACCOUNTING



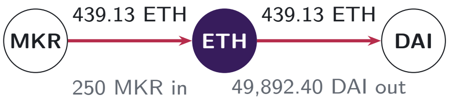
One PoolManager holds all pools. During a lock, the two swaps update balance deltas; the caller settles its obligations before the lock ends.

Where is the route settled?

Economic implication. Pair creation changes available paths; concentrated liquidity changes pair-specific execution at the current state and trade size; the singleton changes the settlement boundary for a route that still uses two pools.

A6. V1 forced routes have an on-chain signature

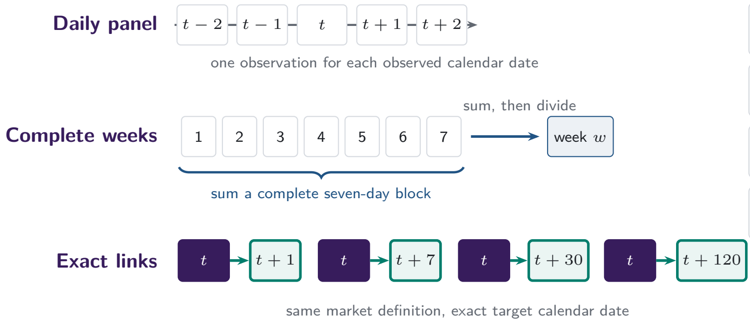
- Token-to-token execution emits two linked swaps through ETH.
- Matching ETH quantities distinguish protocol-imposed intermediation from unrelated bundled swaps.
- 129,810 mandate-era token-to-token transactions carry exactly matching ETH legs; the trace shows the largest.



same transaction, matching ETH flow

Note: Both legs of transaction `0x4dca160d...96a0ca16` (15 March 2020, block 9,674,728) report the same ETH amount to the last recorded digit.

A7. Daily and weekly frequencies answer different questions



Model role

Daily panel date fixed effects

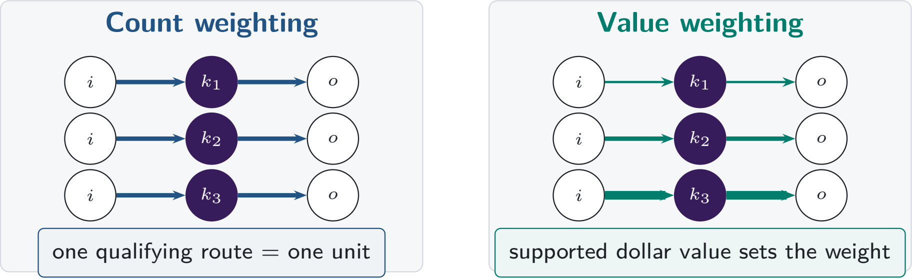
Calendar check week and weekday fixed effects

Complete weeks week fixed effects

Exact links inference by origin date

Missing target dates remain missing: row shifts and calendar months do not substitute for exact calendar links.

A8. Count versus value weighting



Note: Both panels contain the same routes. Line widths are illustrative, and value weighting applies only on the stated valuation-support band.

Vehicle dominance aggregates realised intermediary choices

One realised route

k occupies the intermediary position

Vehicle dominance

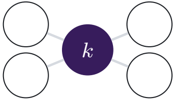
D_k = use through k
$\div$ total intermediary use

Value-weighted routes use comparable dollar amounts

Excess vehicle use

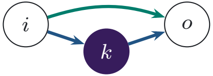
$X_k = D_k \div E_k$
E_k: endpoint share of k
$X_k > 1$: intermediary use exceeds the endpoint benchmark

Separate characteristics of the asset and route set



Network position

position of k in the feasible graph



Execution cost

which route returns more at the same state

Observed route-endpoint use supplies the benchmark; network position and same-state route costs remain separate characteristics.

A9. Intermediary and route-endpoint use are separate

Intermediary share

$$I_{k,t}^{N} = \frac{n_{k,t}^{I}}{\sum_j n_{j,t}^{I}}$$

How often asset k sits inside a route.

Endpoint share

$$E_{k,t}^{N} = \frac{n_{k,t}^{E}}{\sum_j n_{j,t}^{E}}$$

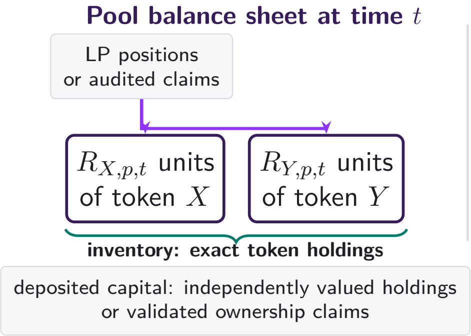
How often asset k appears at either end of the observed pool route.

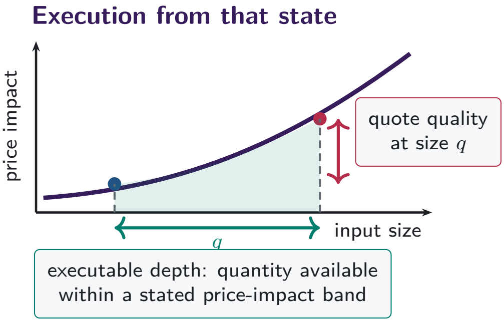
The benchmark describes reconstructed pool routes; an aggregator instruction can begin or end outside the connected component.

Excess use compares intermediation with observed route-endpoint use on the same support.

Note: For USDT in 2024–26, within-category changes account for 93.7% of the count change and 85.6% of the routed-value change; reweighting between single- and cross-venue categories accounts for 6.3% and 14.4%. The identity carries no sampling interval.

A10. Capital, inventory, and depth are different objects

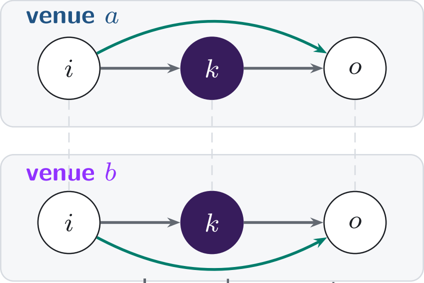




Capital is a stock. Depth and quotes depend on trade size and current pool state.

A11. Additional venues expand the feasible route set

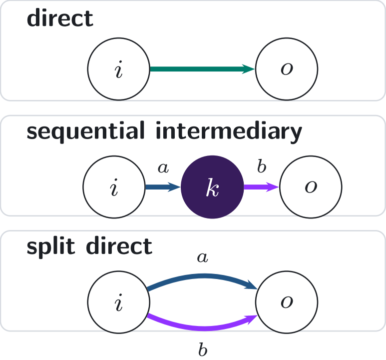
Feasible venue layers



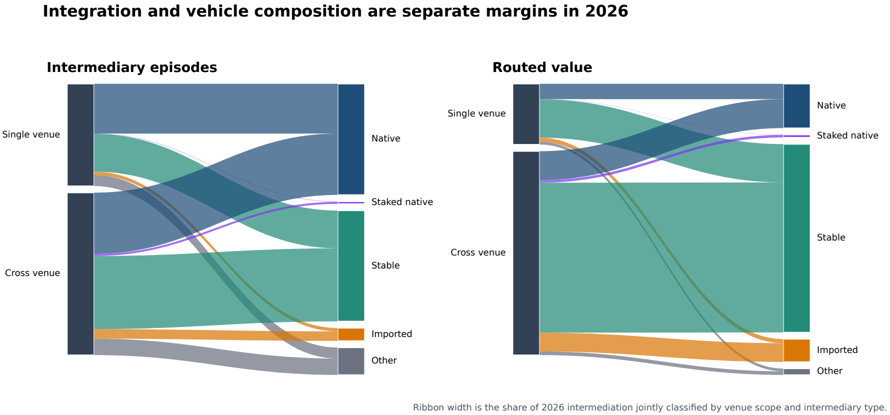
venue layers share assets;
available links define reach

realised
execution

Realised route topology



A11b. Venue scope and vehicle type differ in 2026



Note: Ribbon width is the share of 2026 intermediation jointly classified by venue scope and intermediary type.

A12. References: vehicle currencies and market structure

- Amiti, Itskhoki and Konings (2022), *Quarterly Journal of Economics*
- Dowd and Greenaway (1993), *Economic Journal*
- Flandreau and Jobst (2009), *Economic Journal*
- Gopinath and Stein (2021), *Quarterly Journal of Economics*
- Krugman (1980), *Journal of Money, Credit and Banking*
- Makarov and Schoar (2020), *Journal of Financial Economics*

A13. References: decentralised exchange design

- Adams et al. (2021), *Uniswap v3 Core*
- Adams et al. (2024), *Uniswap v4 Core*
- Angeris, Chitra, Evans and Boyd, optimal routing in constant-function markets
- Barbon and Ranaldo, decentralised exchange costs and market structure
- Lehar and Parlour (2024), *Journal of Finance*
- Milionis, Moallemi, Roughgarden and Zhang, loss versus rebalancing
- Xu, Paruch, Cousaert and Feng, AMM design and slippage